JPEG-LS ENCODER IP CORE · V1.3

OpenJLS

Lossless JPEG-LS encoder for FPGAs.
One pixel per clock, no external memory.

280 MPx/s	1.6:1	8–16 bit	5639/5639
THROUGHPUT, 12-BIT	COMPRESSION	PIXEL DEPTH	BYTE-EXACT IMAGES

01

Overview

OpenJLS is a lossless JPEG-LS encoder IP core for FPGAs. It takes one grayscale pixel per clock over a ready/valid stream and emits a standard JPEG-LS bitstream, buffering a single image row on-chip. It is portable VHDL, verified byte-exact against the CharLS reference in simulation and on silicon.

Source and verification suite: `https://github.com/isentropic-fpga/OpenJLS`
Demo projects: `https://github.com/isentropic-fpga/OpenJLS-Demos`

Table 1 Key specifications.

Standard	JPEG-LS[1]
Compression	Lossless
Compression ratio	≈1.6:1[2]
Components	Grayscale
Pixel depth	8–16 bit[3]
Image size	Up to ≈64k × 64k px[4]
Throughput	1 pixel / clock
Latency	14 cycles
f_{max}	≈280 MHz[5]
Interface	Ready/valid stream[6]
Memory	On-chip only[7]
Resources	5.3–7.6k LUT, 1.8–2.3k FF[8]
Portability	Vendor-agnostic VHDL

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Functional description

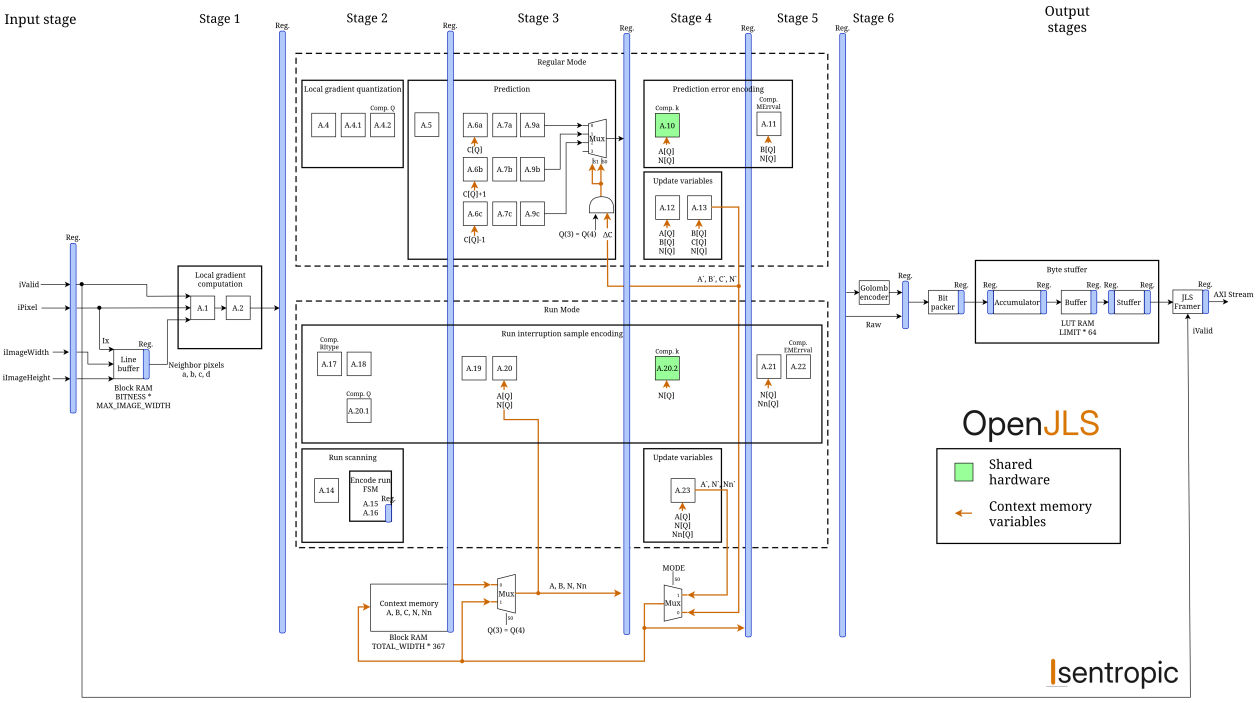
The core is a fully pipelined encoder with a latency of 14 clock cycles: an input register, six processing stages (gradient computation, context modeling, MED prediction with bias correction, Golomb–Rice / run-length coding) and seven internally registered output stages (bit packer, byte stuffer, framer). The architecture builds on Mert [2]; RAM primitives come from open-logic [4].

End-of-image is derived internally from the dimensions provided at the ports `iImageWidth` and `iImageHeight`. The core never stalls at end of line or end of image, so images stream back-to-back. It stalls only when downstream backpressure fills the byte-stuffer buffer. In principle the byte stuffer's fixed 4-byte-per-clock drain could also stall the input if the code rate exceeded it for long enough, but none of the random, adversarial or worst-case natural images we measured caused a stall.

The output stages assert `oValid` only once a full `OUT_WIDTH` word is ready, so output beats are much rarer than input pixels.

[1] ISO/IEC 14495-1, ITU-T T.87.
[2] Aggregate over the 240 natural photographs of the test corpus; content-dependent.
[3] Set by the `BITNESS` generic.
[4] Nominally 65535 × 65535 (64k − 1), the largest value that fits the 2-byte dimension fields.
[5] 12-bit, Xilinx `xczu7eg-1`.
[6] AXI4-Stream and Avalon-ST compatible.
[7] One image row; no external RAM.
[8] Across 8–16 bit, plus block RAM that scales with image width and pixel depth.

Figure 1 Encoder pipeline architecture.



A full-resolution version of this diagram is available in the repository.

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Generics

Table 2 Configuration generics.

GENERIC	RANGE	DESCRIPTION
`BITNESS`	8–16	Pixel bit depth.
`MAX_IMAGE_WIDTH`	4–65535	Largest width supported; sets line-buffer depth.
`MAX_IMAGE_HEIGHT`	1–65535	Largest height supported.
`OUT_WIDTH`	48–1024	Output bus width in bits; multiple of 8.

`MAX_IMAGE_WIDTH` and `MAX_IMAGE_HEIGHT` set the *compile-time* maximum and size the line buffer and dimension counters. Each image's actual size is set at *run time* through `iImageWidth` and `iImageHeight` (Table 3), up to that maximum.

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Ports

All ports are synchronous to the rising edge of `iClk` — the core is a single clock domain. The streaming ports use a plain ready/valid handshake; in Table 3 the **AXIS** column gives the one-to-one AXI4-Stream signal mapping for that ecosystem (Avalon-ST maps the same way at `readyLatency` = 0).

Table 3 Port list and AXI4-Stream mapping.

PORT	DIR	WIDTH	AXIS	ROLE
iClk	in	1	—	Clock.
iRst	in	1	—	Synchronous reset, active high; latches dimensions.
iImageWidth	in	16	—	Image width, pixels (configuration).
iImageHeight	in	16	—	Image height, pixels (configuration).
iValid	in	1	TVALID	Pixel valid.
iPixel	in	BITNESS	TDATA	Pixel data.
oReady	out	1	TREADY	Input ready.
oData	out	OUT_WIDTH	TDATA	Bitstream beat.
oValid	out	1	TVALID	Output valid.
oKeep	out	OUT_WIDTH/8	TKEEP	Valid-byte mask on final beat.
oLast	out	1	TLAST	End of image.
iReady	in	1	TREADY	Downstream ready.

IMPORTANT:

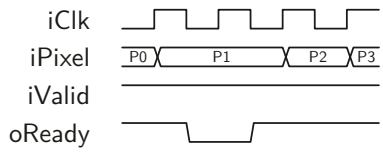
- iImageWidth and iImageHeight are only sampled while iRst is asserted ('1').
- Out-of-range values (width outside 4–MAX_IMAGE_WIDTH, height outside 1–MAX_IMAGE_HEIGHT) fall back to the maximum.
- If you don't intend to change the image dimensions at run-time, wire both inputs to 0 so that the maximum values are used.
- Both ports are a fixed 16 bits regardless of the MAX_IMAGE_* generics, which keeps them usable in Vivado block designs (their port-width evaluator handles only literal arithmetic).

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Timing and protocol

Pixel input. Pixels are streamed in scan order — the first row left to right, then the second row, and so on — each an unsigned value on iPixel. A pixel transfers on every cycle where iValid and oReady are both high. Outside reset, the core holds oReady high and drops it only when the byte-stuffer buffer fills under downstream back-pressure. Figure 2 shows a transfer with a brief downstream stall.

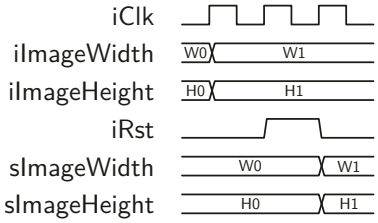
Figure 2 Pixel-input handshake.



No pixel is dropped while oReady is low.

Reset and configuration. iRst is level-sensitive (hold it high for at least one clock) and clears the pipeline. The image dimensions are captured only while iRst is high: present iImageWidth and iImageHeight, hold them stable across the reset pulse, and they latch on the rising edge (Figure 3). A reset is therefore needed only to capture the dimensions — before the first image and whenever the image size changes; with the size unchanged, images encode back-to-back.

Figure 3 Reset and dimension capture.



`sImageWidth` and `sImageHeight` are the internal registers that hold the dimensions; they load on the rising edge while `iRst` is high.

Output. Beats appear on `oData`/`oValid` subject to `iReady`; `oLast` marks the final beat (EOI) and `oKeep` flags its valid bytes. The bitstream is byte-serial and MSB-first: the earliest byte occupies the most-significant byte of `oData`.

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Integration example

```
u_enc : entity work.openjls_top
  generic map (
    BITNESS => 8,
    MAX_IMAGE_WIDTH => 4096,
    MAX_IMAGE_HEIGHT => 4096,
    OUT_WIDTH => 64 )
  port map (
    iClk => clk,
    iRst => rst,
    iImageWidth => img_w, -- latched on reset
    iImageHeight => img_h, -- latched on reset
    iValid => s_valid, -- pixel stream (AXIS slave)
    iPixel => s_data,
    oReady => s_ready,
    oData => m_tdata, -- bitstream (AXIS master)
    oValid => m_tvalid,
    oKeep => m_tkeep,
    oLast => m_tlast,
    iReady => m_tready );
```

It can also be added to a block design as an RTL module.

Xilinx IP cores

For the Vivado flow the core ships the following pre-packaged IPs:

IP CATALOG NAME	INTERFACES
OpenJLS Encoder (Native)	Plain ready/valid
OpenJLS Encoder (AXI4-Stream)	AXI-Stream for the data, native pins for control
OpenJLS Encoder (AXI4-Stream + AXI4-Lite)	AXI-Stream for the data, AXI-Lite for control

Add `Sources/Xilinx/ip_repo/` to the project's IP repositories and the three cores appear in the IP Catalog, ready to drop onto a block design.

AXI4-Lite register map

The AXI4-Lite variant replaces the native control pins with a register bank (32-bit registers, word-aligned offsets):

OFFSET	NAME	ACCESS	CONTENTS
0x00	ID	RO	ASCII "OJLS" (0x4F4A4C53)
0x04	VERSION	RO	0x00MMmmpp (major/minor/patch)
0x08	CAPS	RO	[7:0] BITNESS, [15:8] output bytes per beat (OUT_WIDTH/8)
0x0C	MAXDIM	RO	[15:0] MAX_IMAGE_WIDTH, [31:16] MAX_IMAGE_HEIGHT
0x10	WIDTH	RW	[15:0] image width (clamped on write)
0x14	HEIGHT	RW	[15:0] image height (clamped on write)
0x18	CTRL	WO	[0] APPLY — self-clearing, pulses the core reset
0x1C	STATUS	RO	[0] BUSY (reset pulse active), [1] pixel-stream TREADY mirror

The core samples the dimensions only while its reset is high, so reconfiguration is: write WIDTH/HEIGHT, set CTRL.APPLY. APPLY pulses the core reset for one clock while the registers are held stable; the AXI-Lite endpoint itself stays on aresetn only, so the bus never disappears mid-reconfigure. While the pulse is active STATUS.BUSY reads 1 and the pixel stream's TREADY is held low, so a stream started too early stalls instead of losing pixels — WIDTH/HEIGHT/CTRL writes while BUSY are dropped. **APPLY aborts any image in flight — only apply between images.** Back-to-back images of unchanged dimensions need no APPLY.

WIDTH/HEIGHT writes are merged per WSTRB and clamped to the core's rule (out-of-range values become the MAX generic), so a readback always returns the value the core will actually use. Writes to RO or unmapped offsets are acknowledged (OKAY) and dropped; unmapped reads return zero.

For driver code there is a C header in the repository, ojls_regs.h.

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Resources and performance

Characterized on Xilinx Zynq UltraScale+ xczu7eg-fbvb900-1-e, Vivado 2025.2, with OUT_WIDTH = 64, RTL-only with no floorplanning. f_{max} is the true maximum, found by over-constraining the clock until timing fails. At one pixel per clock, ≈280 MHz is ≈280 Mpixel/s.

Every configuration is implemented with four strategies: the Vivado default, EPRPO (Performance_Explore PostRoutePhysOpt), NetDelay (Performance_NetDelay_high) and Congestion (Congestion_SpreadLogic_high). Frequencies are in MHz; bold marks the best strategy per row. Resources are for the default strategy and near-identical across strategies.

Image width sweep

BITNESS = 12, MAX_IMAGE_WIDTH from 4096 to 65535. The design is congestion-bound, so the winning strategy shifts with image size and with netlist changes; the best-of band (≈282–289 MHz) is the headline figure. Strategies differ by 7–22 MHz per size, so a default run lands close to it. Logic is essentially constant with image size; only the line-buffer BRAM scales with width.

Figure 4 Maximum frequency vs maximum image width.

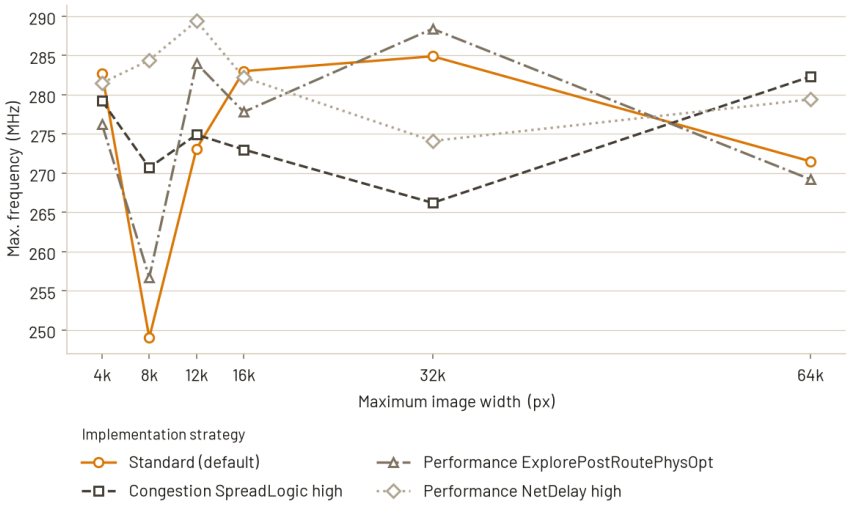


Table 4 Maximum frequency by maximum image width and strategy.

MAX_IMAGE_WIDTH	DEFAULT	EPRPO	NETDELAY	CONGESTION
4096	**282.7**	276.2	281.5	279.2
8192	249.0	256.7	**284.4**	270.7
12288	273.1	284.0	**289.4**	274.9
16384	**283.0**	277.8	282.2	273.0
32768	284.9	**288.4**	274.1	266.2
65535	271.5	269.2	279.4	**282.3**

Figure 5 Resource usage vs maximum image width.

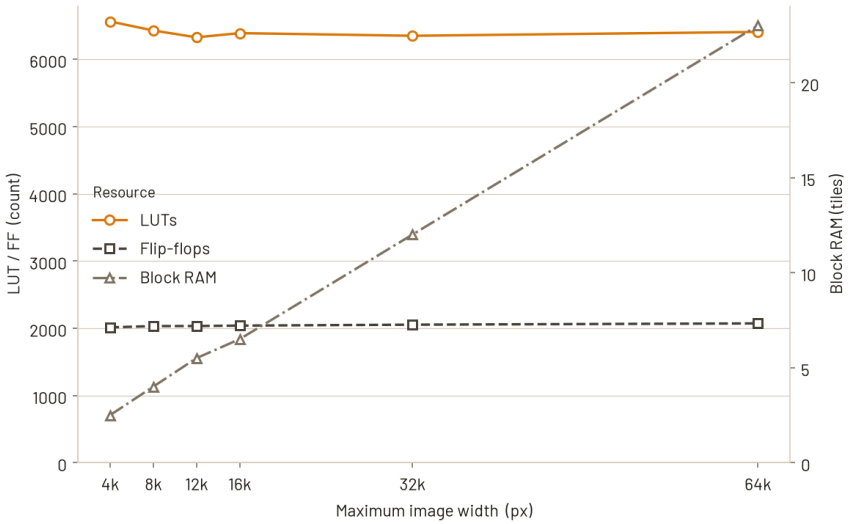


Table 5 Resource usage by maximum image width.

MAX_IMAGE_WIDTH	LUT	FF	BRAM
4096	6563	2013	2.5
8192	6429	2032	4.0
12288	6327	2033	5.5
16384	6389	2039	6.5
32768	6350	2053	12.0
65535	6407	2072	23.0

Pixel depth sweep

`MAX_IMAGE_WIDTH = 12288`, `BITNESS` from 8 to 16. Best-of f_{max} falls from ≈305 MHz at 8 bits to ≈277 MHz at 14 bits and ≈225 MHz at 16 bits. The byte stuffer is off the critical path at every depth; the limit is the context RAM read into the prediction-error register, whose carry chains widen with pixel depth. Logic and line-buffer BRAM both grow with depth.

Figure 6 Maximum frequency vs pixel bit depth (`MAX_IMAGE_WIDTH = 12288`).

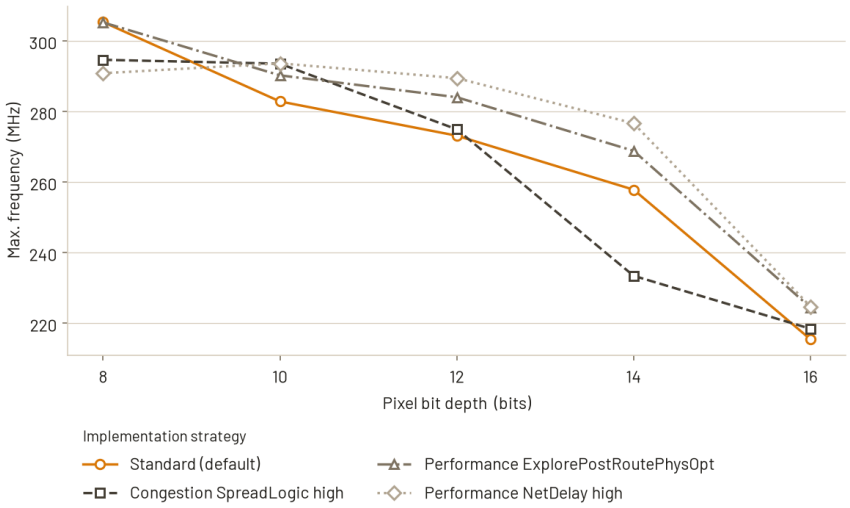


Table 6 Maximum frequency by pixel bit depth and strategy (`MAX_IMAGE_WIDTH = 12288`).

BITNESS	DEFAULT	EPRPO	NETDELAY	CONGESTION
8	**305.3**	305.2	290.8	294.6
10	282.8	290.2	**293.6**	293.5
12	273.1	284.0	**289.4**	274.9
14	257.8	268.8	**276.6**	233.4
16	215.4	224.4	**224.7**	218.4

Figure 7 Resource usage vs pixel bit depth (`MAX_IMAGE_WIDTH = 12288`).

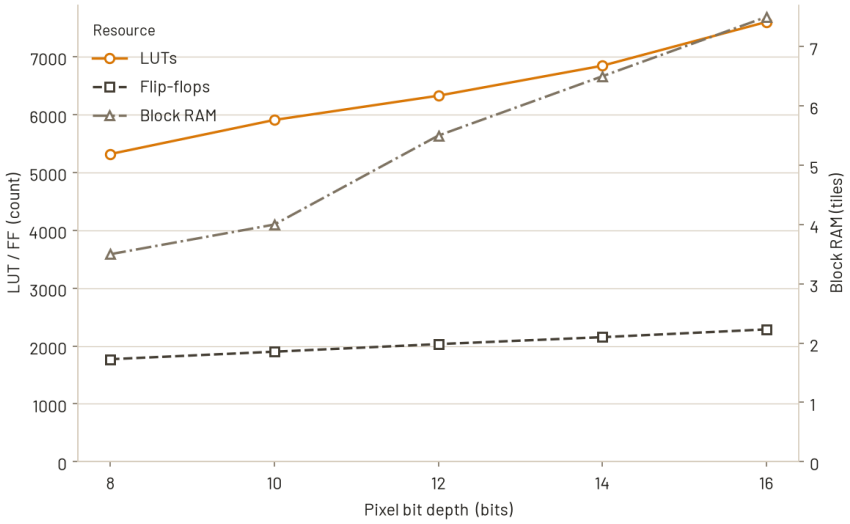


Table 7 Resource usage by pixel bit depth (`MAX_IMAGE_WIDTH = 12288`).

BITNESS	LUT	FF	BRAM
8	5319	1772	3.5
10	5908	1902	4.0
12	6327	2033	5.5
14	6845	2153	6.5
16	7601	2287	7.5

08

Conformance and verification

Verified by simulation with NVC using a layered, self-checking suite run at both RTL and post-synthesis (gate-level) stages:

- **OSVVM** — Per-module tests against independent T.87-derived reference models, top-level control-plane stress (reset injection, backpressure, back-to-back images, dimension fallback), and AXI wrapper tests (byte-exact AXI4-Stream pass-through at 8- and 12-bit, AXI4-Lite register map, live reconfiguration, mid-image abort), all with requirements tracking.
- **Coverage** — OSVVM functional coverage with NVC structural code coverage.
- **Golden model** — Byte-exact vs CharLS on all 5555 simulation-size images (up to 0.5 MP, 8–16 bit) plus the ISO/IEC 14495-1 vectors.
- **Design contracts** — Embedded PSL assertions (ready/valid handshake protocol, internal handshakes) on every run.
- **Post-synthesis** — The stress test and a golden-model subset re-run on the synthesized netlist.
- **Hardware-in-the-loop** — The golden-model comparison on silicon: the full 5639-image corpus, including the 84 full-size originals, encoded on a PYNQ-Z2 FPGA and byte-compared against CharLS. Run by the EncodeOverEthernet project in OpenJLS-Demos.

The golden corpus (Table 8) combines natural photographs (USC-SIPI, imagecompression.info), which exist only at 8 and 16 bits, with natural-derived crops of the 16-bit photographs rescaled to each depth 9–16, seeded white and $1/f^{\beta}$ noise probes, and boundary/adversarial stress probes. Images over 0.5 MP are sliced into full-width strips for simulation; their 84 full-size originals are part of the corpus but run on hardware only.

Table 8 Golden-model corpus per bit depth.

BITS	NATURAL	NATURAL DERIVED	NOISE	STRESS	FULL-SIZE	TOTAL
8	363	0	300	0	69	732
9	0	300	300	2	0	602
10	0	301	300	2	0	603
11	0	300	300	2	0	602
12	0	300	300	2	0	602
13	0	300	300	2	0	602
14	0	301	300	2	0	603
15	0	300	300	2	0	602
16	45	300	300	31	15	691
All	408	2402	2700	45	84	5639

Table 9 Latest regression snapshot.

SUITE	STATUS	TEST/COV	SUMMARY
NVC code coverage	info	99.8%	Statement coverage[9]
OSVVM suite	PASS	100%	113 tests[10]
Golden model	PASS	100%	5555/5555 images[11]
Post-synth OSVVM	PASS	100%	Control-plane stress[12]
Post-synth golden model	PASS	100%	167/167 images[13]
HIL golden model	PASS	100%	5639/5639 images[14]

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Revision and contact

Table 10 Revision history.

VERSION	DATE	NOTES
1.0	2026-06	Initial release.
1.1	2026-07	Packaged IPs for Vivado.
1.2	2026-07	Isentropic branding; cores repackaged under the `isentropic` vendor.
1.3	2026-09	Frequency increased ≈15%, LUT reduction ≈16%, BRAM up 1 tile.

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OpenJLS is developed and maintained by Isentropic. For commercial licensing, technical questions, or collaboration inquiries, contact **contact@isentropic.com.br**.

[9] Per-DUT-file breakdown.
[10] 7 404 577 affirmations (module + top + AXI wrappers).
[11] Byte-exact vs CharLS.
[12] On the gate-level netlist.
[13] 8-bit images up to 0.05 MP, byte-exact vs CharLS.
[14] Byte-exact vs CharLS on PYNQ-Z2 silicon.

References

[1] ITU-T T.87 | ISO/IEC 14495-1, *Lossless and near-lossless compression of continuous-tone still images — Baseline (JPEG-LS)*.
[2] Y. M. Mert, *Key Architectural Optimizations for Hardware Efficient JPEG-LS Encoder*, IEEE, 2018.
[3] CharLS, JPEG-LS reference C++ library, `https://github.com/team-charls/charls`.
[4] open-logic, vendor-agnostic VHDL standard library, `https://github.com/open-logic/open-logic`.